

Particulate Matter Monitoring & Health Effects

Issue 2026-08-18 | Entry window 18 August 2026 | 18 records carried, 12 rejected

18RECORDS CARRIED
(12 REJECTED)**8**SUBTOPIC CLUSTERS
REPRESENTED**9**EFFECT ESTIMATES
WITH INTERVALS**9**INSTRUMENTATION /
EXPOSURE RECORDS**2**METADATA-ONLY
(LABELLED)

Signal of the day

A single *ES&T* issue supplies both halves of the calibration problem this watch tracks — the network-scale fix and the reason the network's own target quantity is the wrong one.

Wang et al. remove the colocation requirement from low-cost sensor calibration. Their framework pairs uncolocated sensors to reference stations by a spatiotemporal selection algorithm, refits monthly per metropolis, and is validated on **over 33 million hourly observations** from the largest US LCS network across ten metropolitan areas. The equity finding is the part that should be read twice: LCS spatial density varies significantly by race/ethnicity and income, with **African American communities the least covered** — so a remote-calibration method that works is a prerequisite for, not a substitute for, fixing where the sensors physically are. Note what the paper does *not* report in its abstract: no R^2 , no RMSE, no bias figure, only “good agreement”. A calibration paper whose headline claim is agreement and which declines to quantify it cannot be benchmarked against the colocated literature, and this digest will not assert a performance number it was not given.

Against that, Zhang et al. put a mask-format sampler on **12 volunteers** in subway environments and collect inhaled and exhaled aerosol simultaneously. Roughly **66% of inhaled PM is deposited**, and deposition is highest below **1.0 μm** . Every ambient network in the first paper — and every regulatory station it is calibrated against — reports $\text{PM}_{2.5}$ mass, a metric to which the sub-micron fraction contributes little while carrying most of the deposited dose. The two papers are not in tension; they mark the two ends of a gap. We are getting much better at making a dense network agree with a reference instrument, and no better at making that reference quantity the one that reaches the lung.

What changed today

- ▶ **The strongest causal-design evidence of the watch to date lands on a clean air zone.** Wood et al. (*Lancet Public Health*) recruited **3,414 children** across 84 schools in London and Luton *before* ULEZ implementation and followed FEV_1 growth in a parallel-cohort design. London children gained **10 mL/year more** (**95% CI 5–15**, $p = 0.0002$) than Luton children, against a faster NO_2 decline (-3.77 vs $-1.77 \mu\text{g}/\text{m}^3$ per year), and a baseline London deficit of -38 mL reached parity after four years. The design is the point: a contemporaneous control city converts a policy rollout into something close to a natural experiment. The limitation is that the exposure contrast is carried by NO_2 , the traffic-exhaust marker, not by PM — the paper cannot separate the two.
- ▶ **Two independent records argue that $\text{PM}_{2.5}$ mass is the wrong exposure metric, from opposite directions.** Parmar et al. resolve Delhi $\text{PM}_{2.5}$ into components and find **black carbon at 11.5% (3.3–20.4) per $10 \mu\text{g}/\text{m}^3$** against **0.93% (0.10–1.76) for total mass** — an order-of-magnitude difference in per-unit potency. Li et al. (*ES&T*) reach the same place from the bench, identifying chromium and specific condensed water-soluble organic constituents as the fractions that actually cross a human blood-brain-barrier platform. Component-resolving instrumentation is being asked for by both literatures simultaneously.
- ▶ **A null worth carrying.** Zewdie et al. find wildfire smoke $\text{PM}_{2.5}$ *not* associated with tract-level cancer history prevalence across the contiguous US. The two surviving regional signals are small (Midwest **PR 1.020**, Northeast **PR 1.040**) and sit in the *least-smoke* regions — the pattern residual confounding produces, as the authors themselves note.
- ▶ **A widely-embedded aerosol-software default is shown to be wrong most of the time it is used.** Rapp et al. (*AMT*) fit a ten-year Storm Peak size-distribution record with a template-free lognormal decomposition and find **four or more modes required for 55.7% of new-particle-formation data** — against the tri-modal assumption standard in PSD fitting. *AMT* is not PubMed-indexed and Crossref-by-ISSN carried it, as it has carried the sensing block on most recent issues.
- ▶ **Europe PMC contributed nothing, and it is indexing lag, not a broken leg.** A control query returns **3 records of any topic** created on 18 August against 2,794 on 17 August — an argument for backfilling a missed day rather than recording a null.

Corpus analytics

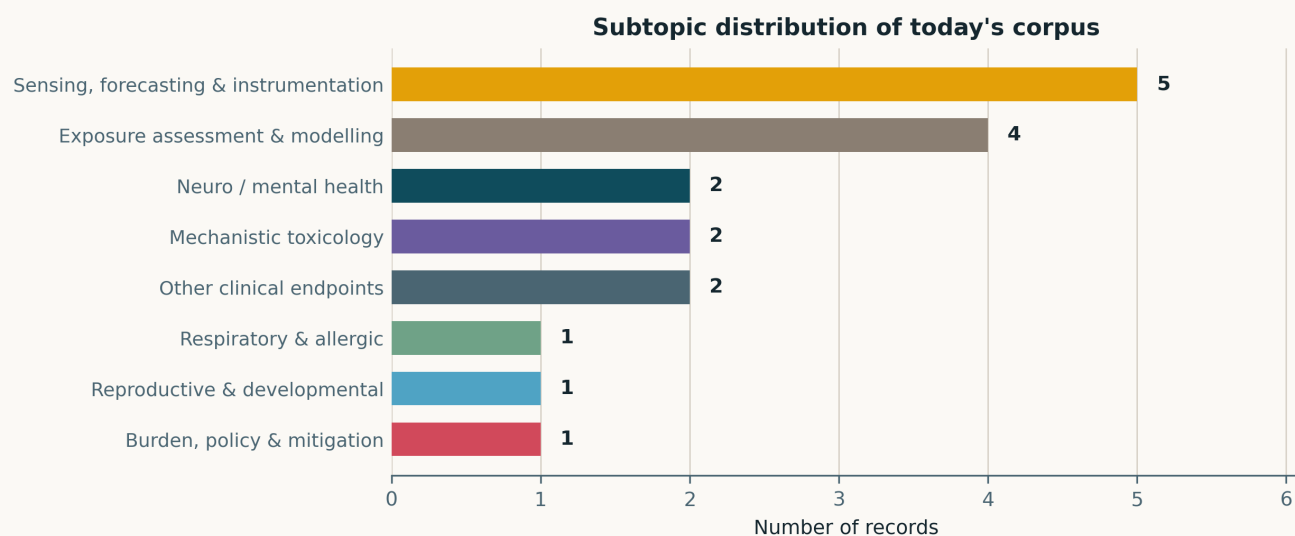


Fig. Subtopic assignment of the 18 carried records, single-label by dominant contribution. *Sensing, forecasting & instrumentation* (5) and *Exposure assessment & modelling* (4) together account for exactly half the issue; that share has now been at or above 0.50 in seven of the last twelve issues, so the instrumentation weighting is a standing feature of this watch rather than a property of today. Eight of ten clusters are populated. *Cardiovascular & metabolic* and *Occupational & indoor* are both empty — the third issue in which that pairing is simultaneously absent, after 4 and 16 August.

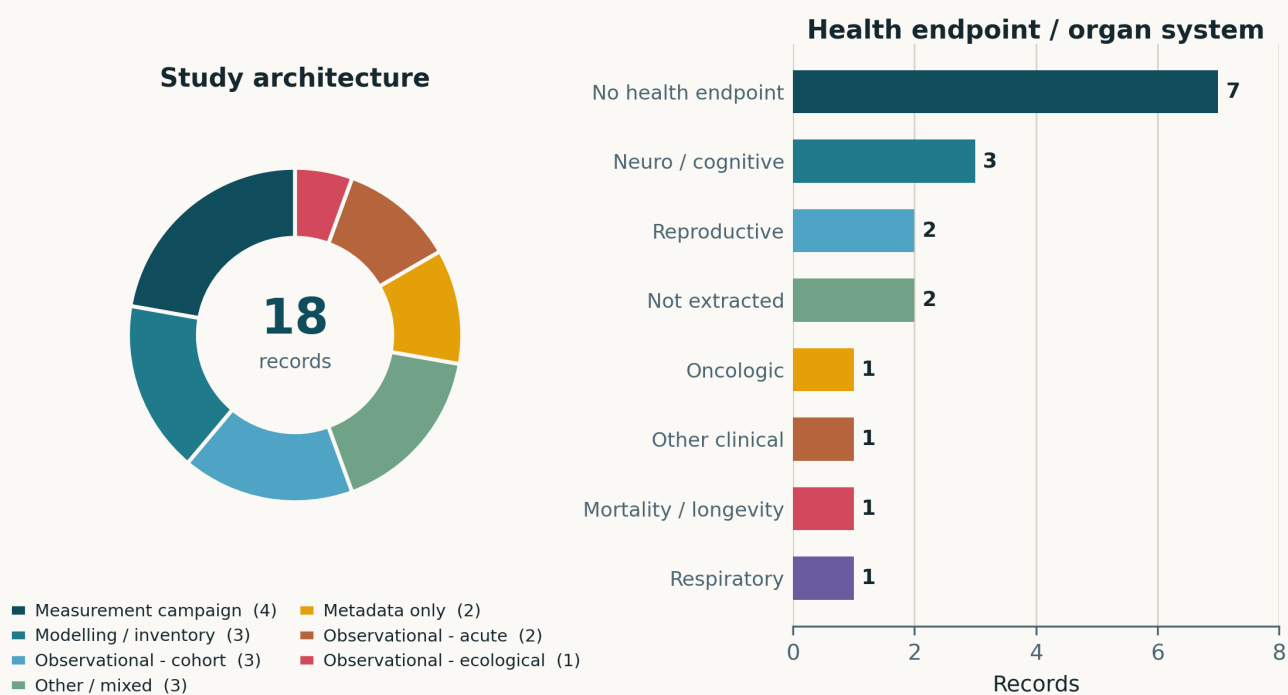


Fig. Left — study architecture. No single design dominates: prospective cohorts (3) and machine-learning models (3) lead, and the tail is long, which is what a corpus split evenly between epidemiology and instrumentation looks like. Right — health endpoint by organ system. Seven records carry an explicit no-health-endpoint label — the direct consequence of the instrumentation share noted above — and a further two are the abstract-withheld records, whose endpoint is unknown rather than absent.

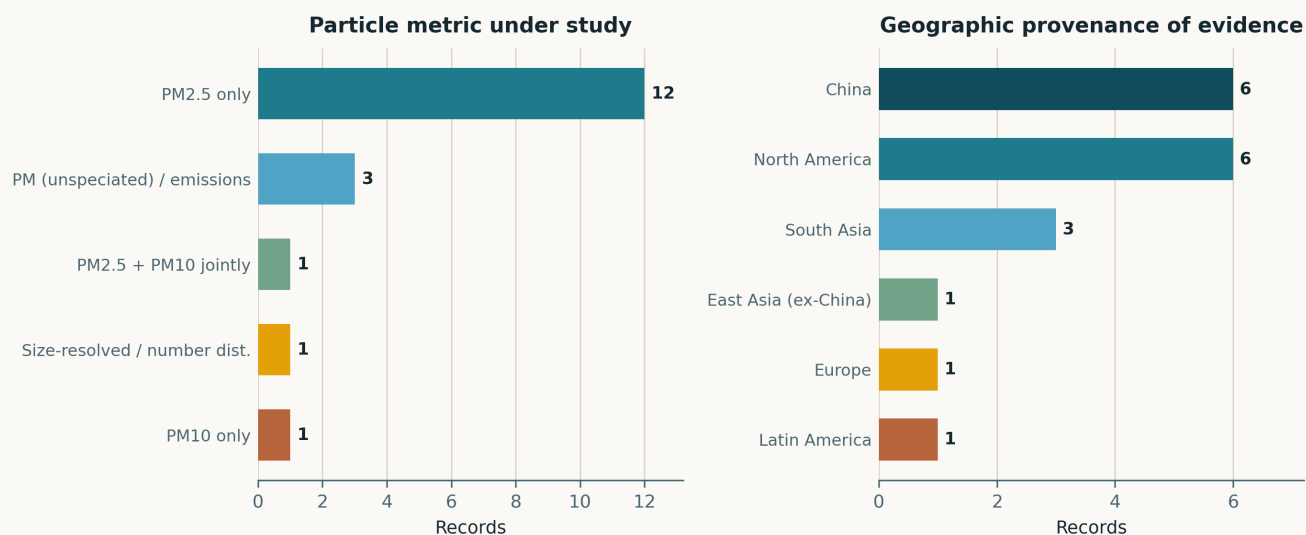


Fig. Left — particle metric. Note the two records whose exposure is explicitly *not* PM_{2.5} mass: Zhang et al. resolve deposition by size below 1 μm , and Rapp et al. work in number-size-distribution space entirely. Right — geographic provenance. The US and China tie at six records each; South Asia contributes two (Delhi, Bangladesh) and Latin America one, the first Argentine record of the watch — which required adding an *Argentina* key and needle to the geography map, without which it would have fallen silently to the unmapped bucket.

Life-course windows addressed today (records may span several stages)

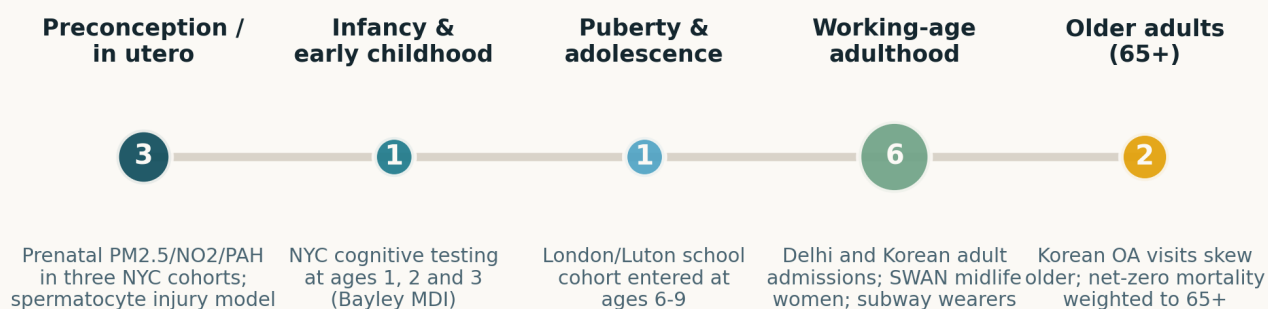


Fig. Life-course windows addressed; counts are non-exclusive. Working-age adulthood dominates (6), carried by the Delhi and Korean admission series, the SWAN midlife cohort and the 12 subway volunteers. The prenatal window (3) ties 15 August for the highest count of the last twelve issues, but the composition differs: here it is one prenatal-exposure cohort plus a cell-line spermatocyte model, not the block of reproductive birth cohorts that drove the mid-August run.

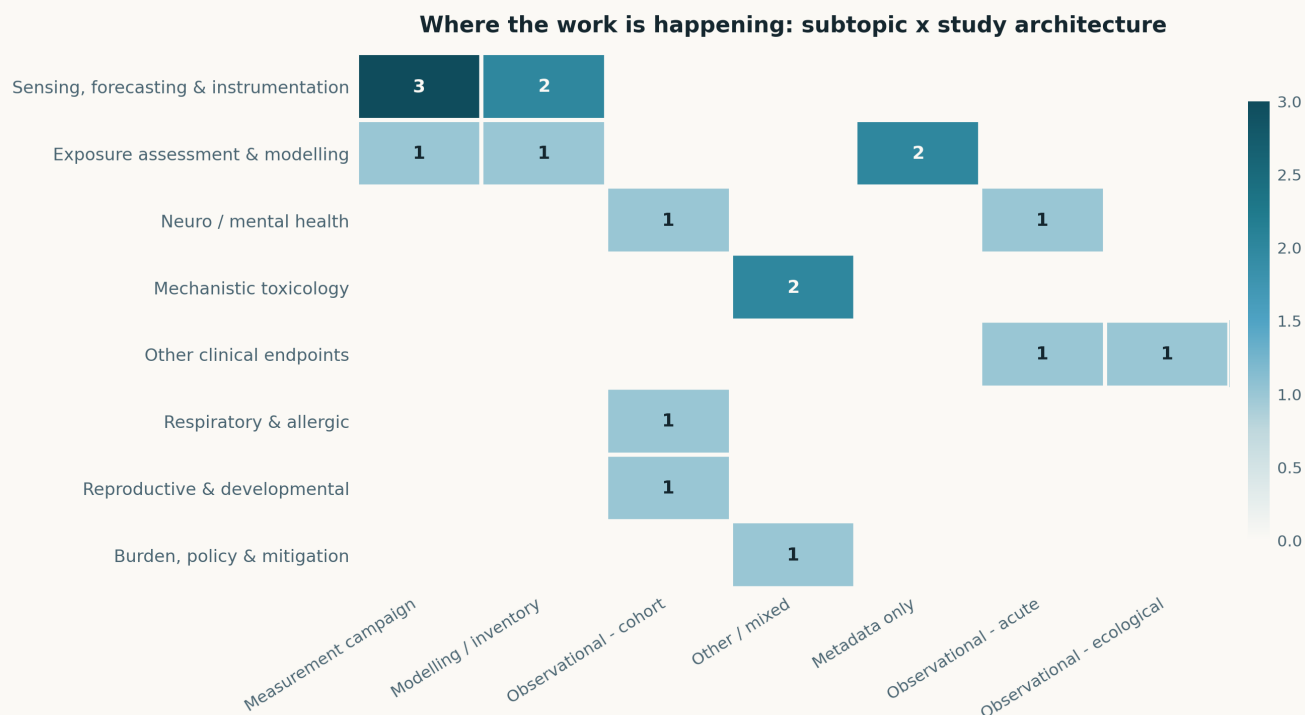


Fig. Subtopic $\times$ study-architecture cross-tabulation. The structural feature of this issue is visible as an empty region rather than a full one: the *Sensing* and *Exposure* rows contain no design that carries a health endpoint. Running the converse check finds a single exception — Perera et al. monitored airborne PAH personally within the CCCEH cohorts, and is the only health-endpoint record here whose exposure was measured by the study team rather than taken from a dispersion model, a reanalysis or a regulatory network. The wearable sampler of Zhang et al. closes that gap instrumentally but carries no health outcome.

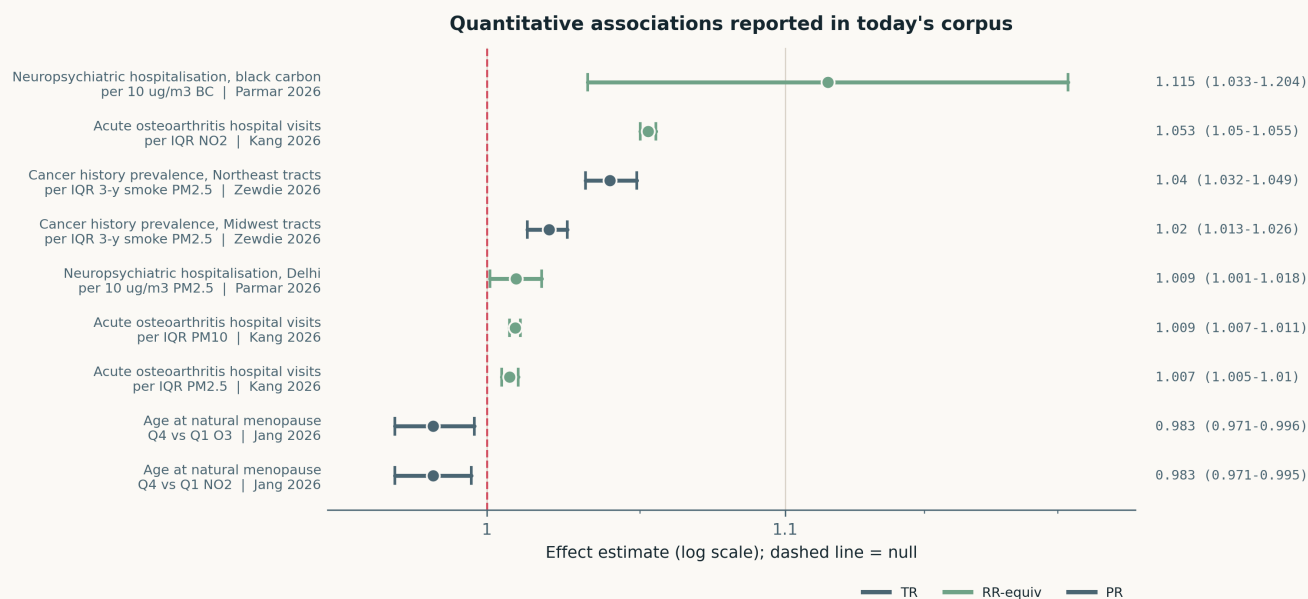


Fig. The nine interval-bearing estimates in today's corpus on a common logarithmic axis. **Exposure contrasts are not mutually comparable:** the Kang et al. and Parmar et al. entries are percentage changes converted to risk-ratio equivalents (per IQR and per 10 $\mu\text{g}/\text{m}^3$ respectively), Zewdie et al. report prevalence ratios per IQR, and the Jang et al. entries are accelerated-failure-time *time ratios* for a top-versus-bottom quartile contrast — these last are below the null by construction and a value under 1 means *earlier* menopause, not lower risk. Read direction and precision only. The tightest intervals belong to the two largest administrative series (3.7 million osteoarthritis visits; all US census tracts) and the widest to the component-resolved black-carbon estimate, which is the one with the smallest effective exposure contrast.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

5 records

Wang et al. 2026 Environ Sci Technol

Remote calibration of a nationwide dense low-cost sensor network for hyperlocal air surveillance

Removes physical colocation from the LCS calibration loop. A spatiotemporal pairing algorithm selects representative training data linking uncolocated sensors to reference stations, with metropolis-specific monthly model refresh; validated on **>33 million hourly observations** across ten US metros. Demonstrated downstream on high-granularity land-use regression and wildfire plume tracking. Two things to hold against it. First, the abstract reports agreement qualitatively and gives *no* R^2 , RMSE or bias, so it cannot be placed against the colocated calibration literature without the full text. Second, and more usefully, the paper quantifies its own denominator problem: LCS density differs significantly by race/ethnicity and income, with African American communities least covered. Remote calibration widens the usable network; it does not move a sensor into a neighbourhood that has none.

doi: 10.1021/acs.est.6c03270

Zhang et al. 2026 Environ Sci Technol

Wearable device for direct assessment of particle deposition in the human respiratory system

Face-mask-format sampler collecting inhaled and exhaled aerosol simultaneously, so deposition is measured by difference rather than inferred from a lung-deposition model. Bench validation confirmed exhaled-particle capture with minimal ambient interference; deployed on **12 healthy volunteers** in subway environments. **~66% of inhaled PM deposited**, highest for PM_{10} ; Fe, Cu and Pb dominated the subway aerosol and deposition varied with lung function and exercise habit. Twelve subjects in one microenvironment is a feasibility study, not a dosimetry dataset, and the size-resolved difference method is vulnerable to hygroscopic growth of exhaled particles at 100% RH — which the abstract does not address. The device concept is nonetheless the most directly relevant instrumentation record of the issue: it measures the delivered dose that every ambient network is a proxy for.

doi: 10.1021/acs.est.6c05543

Rapp et al. 2026 Atmos Meas Tech

Curve fitting algorithm for multimodal particle size distributions: a theoretical basis

Open-source lognormal decomposition with no predefined mode templates and no mode-count constraint. Each mode's N_i , σ_g and $\bar{D}_{pg}$ are fitted by Levenberg–Marquardt and iteratively subtracted until convergence. Evaluated against a ten-year ambient record at Storm Peak Laboratory and against laboratory PSDs of size-selected ammonium sulfate coated in oxidised biogenic SOA. The result that matters: **a tri-modal assumption is inadequate for 55.7% of new-particle-formation data**, four or more modes being required — so the default configuration of most PSD fitting software is wrong in the majority of the regime it is applied to. Agreement with existing models at a tri-modal setting is strong with substantially lower computation. Supplemental routines format output from common mobility particle size spectrometers, which lowers the adoption cost considerably.

doi: 10.5194/amt-19-5373-2026

Li et al. 2026 Environ Sci Technol

Crowdsourced, community-engaged VOC monitoring and source apportionment near a landfill

Community citizen-science campaign near the Bristol, Virginia solid-waste landfill using a modified ultrasonic personal air sampler under EPA Method TO-17. BTEX remained below EPA reference concentrations throughout but was significantly elevated relative to comparison locations. Gas-phase, not particulate — carried here for the deployment methodology rather than the analyte. The transferable content is the study-design integration of resident-reported odour events into sampling triggers, which is the same problem a community PM network faces when episodic sources drive the exposure and a fixed duty cycle misses them.

doi: 10.1021/acs.est.6c04657

Saravagna Sai Lakshmi et al. 2026 J Vis Exp

CORTA-Net: hybrid deep learning with attention-based interpretability for $PM_{2.5}$ forecasting

CorrXGBoost-Rank feature screening, transfer-learning LSTM encoder, multi-head attention for feature- and timestep-level attribution, over CPCB/DPCC station data, IMD meteorology and MODIS fire counts for Delhi 2012–2024. Reported to reduce prediction error against random forest, XGBoost, LSTM and attention-LSTM baselines — but **no RMSE, MAE or R^2 appears in the abstract for either the model or the baselines**, so the size of the improvement is unstated and unverifiable here. A methods-video venue rather than a modelling journal, and the contribution is an assembly of existing components. Included for the fire-count integration, which is the one input choice that is specific to the Delhi problem rather than generic.

doi: 10.3791/71004

Exposure assessment & modelling

4 records

Dias et al. 2026 Bull Environ Contam Toxicol

Micro-scale spatial patterns of $PM_{2.5}$ across urban green and non-green areas in an arid city

Six mobile monitoring campaigns along a fixed 4.3 km transect with **75 georeferenced sampling points** in San Juan, Argentina, winter 2026, classified green versus non-green by surrounding vegetation cover. Generalised linear mixed models adjusting for traffic intensity, distance to nearest road and repeated measurement give **~31% lower adjusted $PM_{2.5}$ in green spaces**. The adjustment set is the weak point — traffic intensity and distance-to-road are the obvious confounders but green spaces in an arid city also differ in surface dust availability and irrigation, which is an unadjusted pathway that would produce the same sign. A single winter and a single transect. Useful mainly as a demonstration that mobile monitoring resolves within-block gradients that a regulatory network cannot, in a South American arid setting where almost no such data exist.

doi: 10.1007/s00128-026-04318-8

Jin et al. 2026 Atmos Pollut Res*Hourly 2 km Himawari-8 aerosol fine-mode fraction retrieval improves PM_{2.5} mapping*

Metadata only. Elsevier withheld the abstract and the record is too recent for Semantic Scholar (404) or Scholar Gateway, whose corpus currently ends in May 2026. The registered title asserts that adding a fine-mode-fraction retrieval to Himawari-8 geostationary observations improves surface PM_{2.5} estimation over the Beijing–Tianjin–Hebei region at hourly, 2 km resolution. **No performance figure, validation design or comparison baseline is asserted here, because none was available.** Flagged for retrieval on a later run. Geostationary fine-mode retrieval is the most direct satellite analogue of the composition-resolution problem raised elsewhere in this issue, so the numbers are worth chasing.

doi: 10.1016/j.apr.2026.103180

Zaman & Mim 2026 Environ Sci Atmos*Decadal variation of atmospheric PM_{2.5} over Bangladesh from MERRA-2 reanalysis*

A 25-year reanalysis climatology of PM_{2.5} and its speciated components across contrasting Bangladeshi microenvironments, presented by the authors as the first comprehensive treatment for a country that ranks among the most PM_{2.5}-polluted globally. The Crossref abstract is truncated at source after the framing sentence, so trend magnitudes, speciation shares and the validation against ground observations are all unavailable here. The methodological caveat is generic but load-bearing for this kind of study: MERRA-2 assimilates AOD, not surface PM, and its skill over South Asia is known to be poorer than over North America or Europe — so a reanalysis-only trend for Bangladesh needs its ground-truth comparison read before the numbers are used.

doi: 10.1039/d6ea00108d

Guo et al. 2026 Atmos Environ*Meteorology-driven long-range transport to an ultra-low emission region during the monsoon*

Metadata only. Elsevier withheld the abstract; Semantic Scholar returns 404 for the DOI. The registered title indicates a Hainan case study of monsoon-driven long-range transport into a low-background region. Nothing beyond the title is asserted. Transport-into-clean-background studies are the useful test case for source attribution because the local emission term is small, which is why this is carried rather than rejected — but it carries no extractable content until the abstract is released.

doi: 10.1016/j.atmosenv.2026.122293

Respiratory & allergic**1 record****Wood et al. 2026** Lancet Public Health*Ultra Low Emission Zone and lung function growth in children: prospective parallel cohort*

The strongest design in the issue. **3,414 children** recruited across 84 schools (44 London, 40 Luton) between June 2018 and April 2019 — *before* ULEZ implementation — with **3,209 (94.0%) in the FEV₁ analysis**. Luton functions as a contemporaneous control city, converting the policy rollout into a difference-in-differences natural experiment rather than a before–after series. Baseline London deficit **–38 mL (95% CI –58 to –18)** against a baseline NO₂ excess of **18.95 µg/m³**; over follow-up London FEV₁ growth exceeded Luton by **10 mL/year (5–15, *p* = 0.0002)** (233 vs 223 mL/year) and mean FEV₁ reached parity at four years. Clinically impaired lung function fell 14%→9% in London and 9%→7% in Luton. Two constraints on reading it as a PM result: the exposure contrast is expressed in NO₂, the exhaust marker, so PM and NO₂ effects are not separable; and two cities differ in more than their air, which parallel-cohort adjustment mitigates but does not eliminate.

doi: 10.1016/S2468-2667(26)00163-5

Neuro, cognition & mental health**2 records****Parmar et al. 2026** Environ Int*Short-term PM_{2.5} and its chemical components and neuropsychiatric hospitalisations*

Ground observations fused with 5 km WRF-Chem output to build a gridded PM_{2.5} field for Delhi 2016–2019, then GAM with distributed lags against hospital neuropsychiatric admissions. Total mass gives **+0.926% (95% CI 0.097–1.762) per 10 µg/m³**; **black carbon gives +11.509% (3.272–20.402)**, ahead of organic carbon and other inorganics, and only industrial-source PM_{2.5} was significant among source contributions. Among disorder categories only anxiety reached significance. The per-unit-mass potency gap between BC and total mass is the headline, but it should be read with the exposure ranges in mind: BC concentrations are roughly an order of magnitude smaller than total mass in Delhi, so a per-10 µg/m³ contrast for BC is a large extrapolation beyond the observed spread. Single-hospital case ascertainment also limits generalisation across a city of Delhi's size.

doi: 10.1016/j.envint.2026.110464

Perera et al. 2026 Environ Health*Declining prenatal air pollution and improving cognitive scores across successive NYC cohorts*

Three successive Columbia CCCEH birth cohorts, 1998–2020, ***n* = 1,109**, with prenatal PM_{2.5}/NO₂ (first two cohorts) and airborne PAH (all three), and Bayley MDI at ages 1, 2 and 3. Exposures fell significantly across cohorts and MDI rose significantly; pooled associations were inverse and significant at all three ages, and cohort-specific inverse associations were clearest in the earliest, highest-exposure cohort (*n* > 700) and weaker in the 2008–2016 cohort. The design is the interesting part — using secular exposure decline as the contrast — and also the vulnerability: everything else about being a child in New York changed between 1998 and 2020, and period effects on Bayley scores are not identifiable from cohort effects here. The authors frame it as policy-relevant evidence; it is better read as consistency between two trends than as an estimated effect.

doi: 10.1186/s12940-026-01328-1

Reproductive & developmental**1 record****Jang et al. 2026** J Clin Endocrinol Metab*Ambient air pollution and age at natural menopause in a multi-ethnic US cohort*

SWAN, **1,457 midlife women** across six US sites 1999–2015, five racial/ethnic groups, with 1 km² annual PM_{2.5}, NO₂ and O₃ linked to residence and age at natural menopause modelled as time-to-event by accelerated failure time. **NO₂ TR 0.983 (0.971–0.995), O₃ TR 0.983 (0.971–0.996)** for the highest versus lowest exposure quartile — roughly one year earlier menopause, which the authors benchmark against the 0.8-year advance seen in current smokers. **PM_{2.5} was not significantly associated**, which is the finding this watch should record plainly rather than fold into a general air-pollution claim. Residential address as the exposure surrogate over a 16-year window in a mobile midlife population is the main measurement limitation.

doi: 10.1210/clinem/dgag340

Mechanistic toxicology

2 records

Li et al. 2026b Environ Sci Technol

Component-resolved PM_{2.5} translocation across a human blood–brain-barrier immune platform

Microfluidic BBB–immune microenvironment platform with compartment-resolved quantification of which PM_{2.5} constituents actually cross, rather than inference from downstream effect. High doses increased barrier permeability and drove constituent translocation; translocated material produced astrocyte reactivity, microglial M1 polarisation, elevated ROS, neuroinflammation and lipid-metabolic dysregulation. Component-resolved correlation implicates **chromium** and the water-soluble organic matter fraction — specifically the condensed species C₂₈H₇O₂N and C₂₁H₁₆N₂SP₂. The dose is described only as “high”, with no concentration given in the abstract and therefore no way to judge ambient relevance, which is the standard failure mode of this literature. The compartment-resolved measurement of translocation is nonetheless a genuine methodological advance over effect-only designs.

doi: 10.1021/acs.est.6c01030

Jin & Zhang 2026 Reprod Toxicol

Astragaloside IV protects against PM_{2.5}-induced spermatocyte injury via cGAS-STING/Nrf2/HMOX1

Network pharmacology and GEO reanalysis (GSE189187) to nominate targets, molecular docking, then validation in GC-2 spermatocytes. PM_{2.5} at 200 µg/mL raised ROS 5.35-fold and apoptosis to 42.83%; AS-IV co-treatment restored viability to 83.3%, suppressed cGAS/p-TBK1/p-IRF3 and activated Nrf2/HMOX1, reducing apoptosis to 18.27%. The docking affinities exceed the reference inhibitors G150 and H-151, which is presented as validation and is not — docking scores do not rank cellular potency. The exposure is a cell-culture suspension four orders of magnitude above any inhaled concentration, and AUC = 1.000 for a single gene in a public transcriptomic dataset is a small-sample artefact rather than a discriminative result. Carried at tier C for the pathway hypothesis only.

doi: 10.1016/j.reprotox.2026.109328

Burden, policy & mitigation

1 record

Liu et al. 2026 Environ Sci Technol

Disproportionate effects of direct air capture on the regional health cobenefits of net-zero

Grid-level (36 km²) air-pollution mortality assessment along China's net-zero CO₂ pathway 2020–2060, contrasting scenarios with and without large-scale DACCS deployment. DACCS substitutes for fossil-fuel phase-out and therefore forgoes the co-pollutant reduction that would otherwise occur: **1.75 million additional cumulative premature deaths**, with the losses concentrated in developing provinces (**Henan –0.13 million and Hebei –0.12 million avoided deaths**) via more active emissions trading. The paradox the authors report honestly is that inter-regional mortality-rate inequality narrows while the distribution of forgone benefit widens. Scenario-model output over a 40-year horizon, so the intervals are structural-uncertainty bounded rather than statistical, and the result is a property of the assumed trading architecture as much as of the technology.

doi: 10.1021/acs.est.6c04728

Other clinical endpoints

2 records

Kang et al. 2026 Environ Pollut

Short-term ambient air pollution and acute osteoarthritis morbidity across subgroups

Time-stratified case-crossover over **3,697,314 OA-related hospital visits** (ICD-10 M15–M19) from the Korean NHIS-NSC 2002–2019, linked to national pollutant fields, with health-screening data enabling individual-level effect modification. Per IQR: NO₂ +5.28% (5.02–5.54), CO +2.88%, SO₂ +1.36%, PM₁₀ +0.90% (0.73–1.07), O₃ +0.76%, PM_{2.5} +0.73% (0.47–1.00) — so the particulate terms are the weakest of the six and NO₂ is the only one robust in two-pollutant models. Enormous power makes intervals tight enough to exclude the null on effects this small, which is exactly when outcome-definition error matters: the authors state plainly that claims-based visit dates index healthcare utilisation, not symptom onset. A same-day traffic-pollution association with a discretionary outpatient visit is at least as consistent with behaviour as with pathophysiology.

doi: 10.1016/j.envpol.2026.128985

Zewdie et al. 2026 Cancer Epidemiol

Wildfire smoke PM_{2.5} and cancer history prevalence at census-tract level

Daily wildfire-specific smoke PM_{2.5} aggregated to cumulative (2006–2020) and recent (2018–2020) means, linked to 2022 tract-level cancer history prevalence for all contiguous-US tracts; quasi-Poisson GAMs with random effects and spatial splines, adjusted for sociodemographics, behaviours, health status, density, climate and region. **No association in fully adjusted models for either exposure window.** Regional effect modification produced small positive estimates in the Midwest (PR 1.020, 1.013–1.026) and Northeast (PR 1.040, 1.032–1.049) — the two regions with the least smoke exposure, a spatial pattern more consistent with residual confounding than with dose. Cross-sectional and ecological, with prevalence of *ever* having cancer set against a 15-year exposure window that cannot precede diagnosis for most cases; latency alone makes this design close to uninformative for incidence. The value here is a well-powered, honestly reported null in a literature prone to amplification.

doi: 10.1016/j.canep.2026.103199

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Parmar et al.	Short-term PM _{2.5} and its chemical components and neuropsychiatric hospitalisations	<i>Environ Int</i>	Time-series	Ground-WRF-Chem fused PM _{2.5} at 5 km, speciated into BC, OC, OIN	India
Perera et al.	Declining prenatal air pollution and improving cognitive scores across successive NYC cohorts	<i>Environ Health</i>	Prospective cohort	Prenatal personal/modelled PM _{2.5} , NO ₂ and airborne PAH	USA
Reproductive & developmental					
Jang et al.	Ambient air pollution and age at natural menopause in a multi-ethnic US cohort	<i>J Clin Endocrinol Metab</i>	Prospective cohort	1 km ² modelled annual PM _{2.5} , NO ₂ and O ₃ at residence	USA
Respiratory & allergic					
Wood et al.	Ultra Low Emission Zone and lung function growth in children: parallel cohort study	<i>Lancet Public Health</i>	Prospective cohort	Modelled residential annual NO ₂ and PM (LAEI dispersion model)	UK
Mechanistic toxicology					
Jin & Zhang	Astragaloside IV protects against PM _{2.5} -induced spermatocyte injury via cGAS-STING/Nrf2/HMOX1	<i>Reprod Toxicol</i>	In vitro exposure	PM _{2.5} suspension at 200 ug/mL in GC-2 spermatocytes	China
Li et al.	Component-resolved PM _{2.5} translocation across a human blood-brain barrier immune platform	<i>Environ Sci Technol</i>	In vitro organ-on-chip	Ambient PM _{2.5} extract, compartment-resolved constituent quantification	China
Sensing, forecasting & instrumentation					
Li et al.	Crowdsourced, community-engaged VOC monitoring and source apportionment near a landfill	<i>Environ Sci Technol</i>	Field measurement	Gas-phase BTEX by EPA TO-17, not PM	USA
Rapp et al.	Curve fitting algorithm for multimodal particle size distributions: a theoretical basis	<i>Atmos Meas Tech</i>	Measurement campaign	Mobility particle size distributions, number-based	USA
Saravagna Sai Lakshmi et al.	CORTA-Net: hybrid deep learning with attention-based interpretability for PM _{2.5} forecasting	<i>J Vis Exp</i>	Machine-learning model	Hourly regulatory-station PM _{2.5} plus MODIS fire counts	India
Wang et al.	Remote calibration of a nationwide dense low-cost sensor network for hyperlocal air surveillance	<i>Environ Sci Technol</i>	Machine-learning model	Low-cost sensor PM _{2.5} vs regulatory reference monitors	USA
Zhang et al.	Wearable device for direct assessment of particle deposition in the human respiratory system	<i>Environ Sci Technol</i>	Field measurement	Paired inhaled and exhaled size-resolved PM, PM ₁ -PM ₁₀	China
Exposure assessment & modelling					
Dias et al.	Micro-scale spatial patterns of PM _{2.5} across urban green and non-green areas in an arid city	<i>Bull Environ Contam Toxicol</i>	Measurement campaign	Mobile-monitored PM _{2.5} along a fixed transect	Argentina
Guo et al.	Meteorology-driven long-range transport to an ultra-low emission region during the monsoon	<i>Atmos Environ</i>	Metadata only (no abstract)	Not extractable from registered metadata	China
Jin et al.	Hourly 2 km Himawari-8 aerosol fine-mode fraction retrieval improves PM _{2.5} mapping	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Satellite AOD fine-mode fraction to surface PM _{2.5}	China
Zaman & Mim	Decadal variation of atmospheric PM _{2.5} over Bangladesh from MERRA-2 reanalysis	<i>Environ Sci Atmos</i>	Machine-learning model	MERRA-2 reanalysis PM _{2.5} and speciated components	Bangladesh
Burden, policy & mitigation					
Liu et al.	Disproportionate effects of direct air capture on regional health cobenefits of net-zero	<i>Environ Sci Technol</i>	Health impact model	Grid-level (36 km ²) modelled PM _{2.5} and O ₃ attributable mortality	China

First author	Focus	Journal	Design	Metric	Region
Other clinical endpoints					
Kang et al.	Short-term ambient air pollution and acute osteoarthritis morbidity across subgroups	<i>Environ Pollut</i>	Case-crossover	District-level PM ₁₀ , PM _{2.5} , NO ₂ , SO ₂ , O ₃ , CO	South Korea
Zewdie et al.	Wildfire smoke PM _{2.5} and cancer history prevalence at census-tract level	<i>Cancer Epidemiol</i>	Ecological panel	Daily modelled wildfire-specific smoke PM _{2.5} , 2006-2020	USA

Provenance and method

Entry window **18 August 2026**, contiguous with the 17 August issue. This issue is a **backfill**: the scheduled 18 August run did not execute, and the window was recovered day-by-day on 19 August before the current day was opened. Sources queried: **PubMed** E-utilities and the PubMed connector on two separate axes (health, 23 records; sensing, 1, already in the health set); **Crossref by journal ISSN** across 18 aerosol and atmospheric titles (7 deposits), which carried the entire instrumentation block — *Atmos. Meas. Tech.*, *Atmos. Chem. Phys.* and *Atmos. Pollut. Res.* are not PubMed-indexed; **Europe PMC** on **CREATION_DATE** (0 records — a control query returns 3 records of *any* topic for 18 August against 2,794 for 17 August, so this is index lag, not harvester failure); **arXiv** under a 3-day recency screen (0 new; newest in-scope posting was 15 August and already carried); **OpenAlex** HTTP 429, sixteenth consecutive failure, no longer retried; **Consensus** returned three 2026 calibration papers, none deposited in-window (ninth consecutive zero-carry, retained as a weekly/monthly gap-filler); **ClinicalTrials.gov** 0 registrations or updates in window, `trials.json` unchanged. **18 records carried, 12 rejected** — rejections were aerosol- or geochemistry-only records with no PM monitoring method and no health endpoint (marine organosulfate formation, iodic-acid nucleation, pyrogenic-carbon radical chemistry, microplastics in snow, brown-carbon and siloxane SOA precursor chemistry, PTR-ToF-MS gas-phase VOC interference, a GHG mapping framework, a vehicle-fleet material-flow model, an OECD renewable-energy panel regression, tropical cloud Doppler measurements and a precipitation trace-metal series). Each is logged with a reason in `state/rejected.jsonl`. Two records ship as **Metadata only (no abstract)**: Elsevier withheld both, Semantic Scholar returned 404, and Scholar Gateway's corpus ends in May 2026. Nothing beyond the registered title is asserted for either. Subtopic, design, particle-metric and geography labels are single-label by dominant contribution and are a judgement, not a controlled vocabulary. Effect estimates are transcribed from abstracts and use heterogeneous exposure contrasts — see the forest-plot caption. All DOIs verified against Crossref. Abstracts remain the copyright of their respective publishers.